Pierre-Louis Filoche

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Professional Objectives & Interests

Seeking opportunities in the development of intelligent systems at the interface between hardware and software, leveraging artificial intelligence and signal/data processing to solve real-world engineering challenges.

Education

École normale supérieure Paris-Saclay, Université Paris-Saclay 2025-2026

- Master 2 - Computer Science
- ***Agrégation de Sciences de l'Ingénieur* (Computer Science) - Ranked 2nd nationwide**
- Courses: Signal and Image Processing, Wavelets, Digital Electronics, High-Performance Computing (SIMD, CUDA), AI (PINNs), Control Engineering

École normale supérieure Paris-Saclay, Université Paris-Saclay 2023–2025

- Master 1 in Electrical Engineering - Digital and Physical Systems
- Bachelor's degree in Electrical Engineering and Computer Science
- Courses: Probabilities, Signal & Image Processing, AI, Statistical Estimation, Optimization, Statistical Physics

Lycée Jean-Baptiste Say, Paris 2021–2023

- Classes Préparatoires PTSI–PT* - intensive scientific program preparing for national engineering competitions
- Focus: Mathematics, Physics, Computer Science, Systems Engineering
- Class Representative

Academic Experience

International Laboratory on Learning Systems (CNRS), Montreal May–July 2025

- Research internship on fundamental artificial intelligence - *Mixture-of-Experts LLMs and expert prediction, NLP*
- Supervision: Loris Marchal and Pablo Piantanida

Supervised Research Work, Laboratoire Méthodes Formelles (ENS) 2024

- AI-based real-time image processing for microorganism characterization in microfluidic systems
- Supervision: Sakina Bensalem & Mélanie Piétri

Project COBRA -ENS 2024

- Designed an autonomous airship able to navigate a predefined trajectory
- Evaluation and implementation of indoor localization techniques (vision and radio-based)

Safran & ENS Hackathon Feb. 2024

- 2nd place - innovative protection system for satellite infrastructures

Extracurricular Involvement

Associative Leadership -ENS 2024–2025

- President -Sailing Association
- Member -Student Council (BDE)

WEIV -Sailing Freshers' Weekend 2024

- Organized a sailing integration weekend for 50 students - €10,000 budget management

Volunteer Youth Worker - LOISIRS PLURIELS 2020

- Supervised children with and without disabilities in an inclusive environment

Skills

- Programming: Python, C, C++ (SIMD and CUDA), Java, JavaScript, HTML/CSS, VHDL, LaTeX
- Languages: English C1 (IELTS), Spanish B1, Chinese A2 (HSK1/HSK2)

Portfolio website